



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to NAVANEE AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

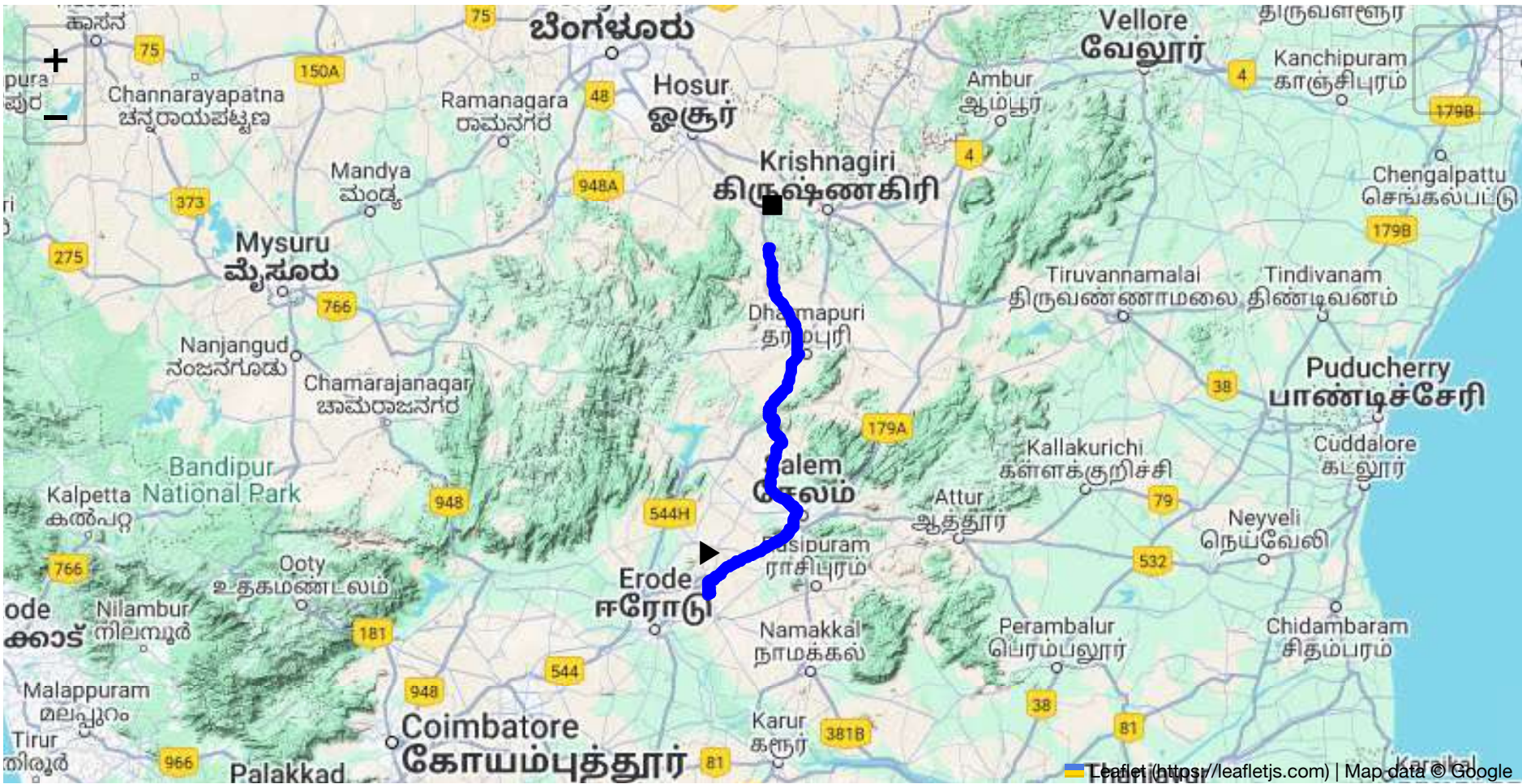
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 140.18 km
Estimated Duration: 2.5 hours
Adjusted Duration (Heavy Vehicle): 3.1 hours

Start: (11.4381, 77.8734)
End: (12.42153, 78.04986)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

- **Starting Point:** CVQF+23W, Sangagiri, Tamil Nadu 637301, India
- **Ending Point:** C3C2+G28, Sokkamatti Rd, Tamil Nadu 636805, India
- **Approximate Distance:** 140.18 kilometers
- **Estimated Travel Time:** Typically 2.51 hours for heavy vehicles with hazardous materials

This route generally involves traveling on national and state highways, known for their mixed traffic with both heavy vehicles and local transport.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather:** Tamil Nadu experiences tropical weather with hot, humid summers (March to May) and a monsoon season (June to September), followed by cooler winters (October to February).
- **Hazards:** Heavy rains during monsoon can lead to waterlogged roads and reduced visibility. High temperatures in summer may cause road fatigue and tire blowouts.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (7:00 AM - 9:00 AM) and evening (5:00 PM - 7:00 PM)
- **Congestion-Prone Areas:**
 - Sangagiri and Erode towns have higher congestion due to local commuter and commercial activity.
 - Intersections with major highways could also experience bottlenecks.

4. Assessment of Road Quality and Infrastructure

- **Highway Sections:** Generally well-maintained but may contain sections with potholes due to weather wear and tear.
- **Local Roads:** Narrower with limited shoulder space, which can be challenging for heavy vehicles.

5. Suggestions for Alternative Routes for Emergencies

- **Alternate Route:** Consider taking NH544 for a section if severe congestion or road closure occurs. It is slightly longer but often less congested.
- **Local Detours:** Utilize local state roads known to have less traffic to bypass specific congested areas.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Transportation of hazardous materials requires adherence to the Central Motor Vehicle Rules.
- Restriction zones typically include urban residential areas; plan to circumvent these as needed.

7. Overview of Historical Incidents

- Past incidents have involved overturning and spillage due to sharp turns and overcrowded road sections.
- Enhanced caution is advised in areas previously recorded for such incidents, especially during adverse weather.

8. Environmental Considerations and Sensitive Areas

- Areas near water bodies and agricultural zones along the route may have stricter environmental protection regulations.
- Special care should be taken to prevent contamination during transit.

9. Analysis of Communication Coverage

- Good coverage in major towns and along highways.
- Potential dead zones in rural sections; using GPS with offline maps recommended.

10. Estimated Emergency Response Times

- **Urban Sections:** Emergency response times typically between 15-30 minutes.
- **Rural Sections:** May extend to 45 minutes to 1 hour due to distance from major towns.

11. Overall Summary of Risk Assessment

This journey entails moderate risk, primarily due to traffic congestion and potential weather disruptions. Key risks include waterlogging during monsoon, high temperatures, and narrow local roads. Proper training on route-specific hazards, adherence to regulations, and consistent communication with dispatch should mitigate most risks. Emergency alternatives and support should be planned ahead, considering the nearest service centers and hospitals along the route. Regular risk assessments and incident reviews are advised to maintain high safety standards for transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.45309, 77.87325	30 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	Medium	11.46765, 77.87037	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.53501, 77.95707	30 KM/Hr
11	Turn	Medium	11.53601, 77.95856	30 KM/Hr
12	Turn	Medium	11.64180, 78.11808	30 KM/Hr
13	Turn	Medium	11.64325, 78.11993	30 KM/Hr
14	Turn	Medium	11.69519, 78.10775	30 KM/Hr
15	Turn	Medium	11.73958, 78.05105	30 KM/Hr
16	Turn	Medium	11.75835, 78.05016	30 KM/Hr
17	Turn	Medium	11.87942, 78.08197	30 KM/Hr
18	Turn	Medium	11.92585, 78.06507	30 KM/Hr
19	Turn	Medium	11.93704, 78.05402	30 KM/Hr
20	Turn	Medium	11.95818, 78.04978	30 KM/Hr
21	Turn	Medium	11.95876, 78.04990	30 KM/Hr
22	Turn	Medium	11.95988, 78.04948	30 KM/Hr
23	Turn	Medium	11.96304, 78.05136	30 KM/Hr
24	Turn	Medium	11.96399, 78.05135	30 KM/Hr
25	Turn	Medium	11.96458, 78.05084	30 KM/Hr
26	Turn	Medium	11.96496, 78.04988	30 KM/Hr
27	Turn	High	11.96580, 78.04923	15 KM/Hr
28	Turn	Medium	11.96675, 78.04939	30 KM/Hr
29	Turn	Medium	11.96791, 78.05070	30 KM/Hr
30	Turn	Medium	12.12058, 78.13283	30 KM/Hr
31	Blind Spot	Blind Spot	12.41986, 78.04639	10 KM/Hr
32	Turn	High	12.41674, 78.04642	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
33	Turn	Medium	12.41625, 78.04783	30 KM/Hr
34	Turn	High	12.41646, 78.04821	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
20	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
21	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
22	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
24	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
25	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
26	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
27	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
28	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
31	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
32	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
33	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
34	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
35	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
29	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
30	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
36	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64325, 78.11993



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69519, 78.10775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73958, 78.05105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75835, 78.05016



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95876, 78.04990



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96458, 78.05084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96496, 78.04988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12058, 78.13283



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.41674, 78.04642

